Engineering Mechanics: Statics

14th Edition
SI Units

Chapter 1 General Principles

Chapter Objectives

- To provide an introduction to the basic quantities and idealizations of mechanics.
- To give a statement of Newton's Laws of Motion and Gravitation.
- To review the principles for applying the SI system of units.
- To examine the standard procedures for performing numerical calculations.
- To present a general guide for solving problems.

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Chapter Outline

1.1 Mechanics

1.2 Fundamental Concepts

1.3 The International System of Units

1.4 Numerical Calculations

1.5 General Procedure for Analysis

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1.1 Mechanics

- Mechanics can be divided into 3 branches:
  - Rigid-body Mechanics
  - Deformable-body Mechanics
  - Fluid Mechanics

- Rigid-body Mechanics is divided into:
  - Statics
  - Dynamics

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1.1 Mechanics

- Statics – Equilibrium of bodies
  - At rest
  - Move with constant velocity

- Dynamics – Accelerated motion of bodies

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1.1 Mechanics

Historical Development

Earlier

Statics

Later

Dynamics

Principles can be formulated simply from measurements of geometry and force

Depend on an accurate measurement of time.

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1.2 Fundamentals Concepts

Basic Quantities

- **Length**
  – locate the position of a point in space
- **Time**
  – conceive as succession of events
- **Mass**
  – measure of a quantity of matter

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1.2 Fundamentals Concepts

Basic Quantities

- **Force**
  - “push” or “pull” exerted by one body on another
  - Occur due to direct contact between bodies
    Eg: Person pushing against the wall
  - Occur through a distance without direct contact
    Eg: Gravitational, electrical and magnetic forces

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1.2 Fundamentals Concepts

Idealizations

- **Particles**
  - Consider mass but neglect size

    Eg: Size of Earth insignificant compared to its size of orbit

- **Rigid Body**
  - Combination of large number of particles
  - Neglect material properties

    Eg: Deformations in structures, machines and mechanism

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1.2 Fundamentals Concepts

Idealizations

- **Concentrated Force**

  – Effect of loading, assumed to act at a point on a body

  – Represented by a concentrated force, provided loading area is small compared to overall size

  Eg: Contact force between wheel and ground

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1.2 Fundamentals Concepts

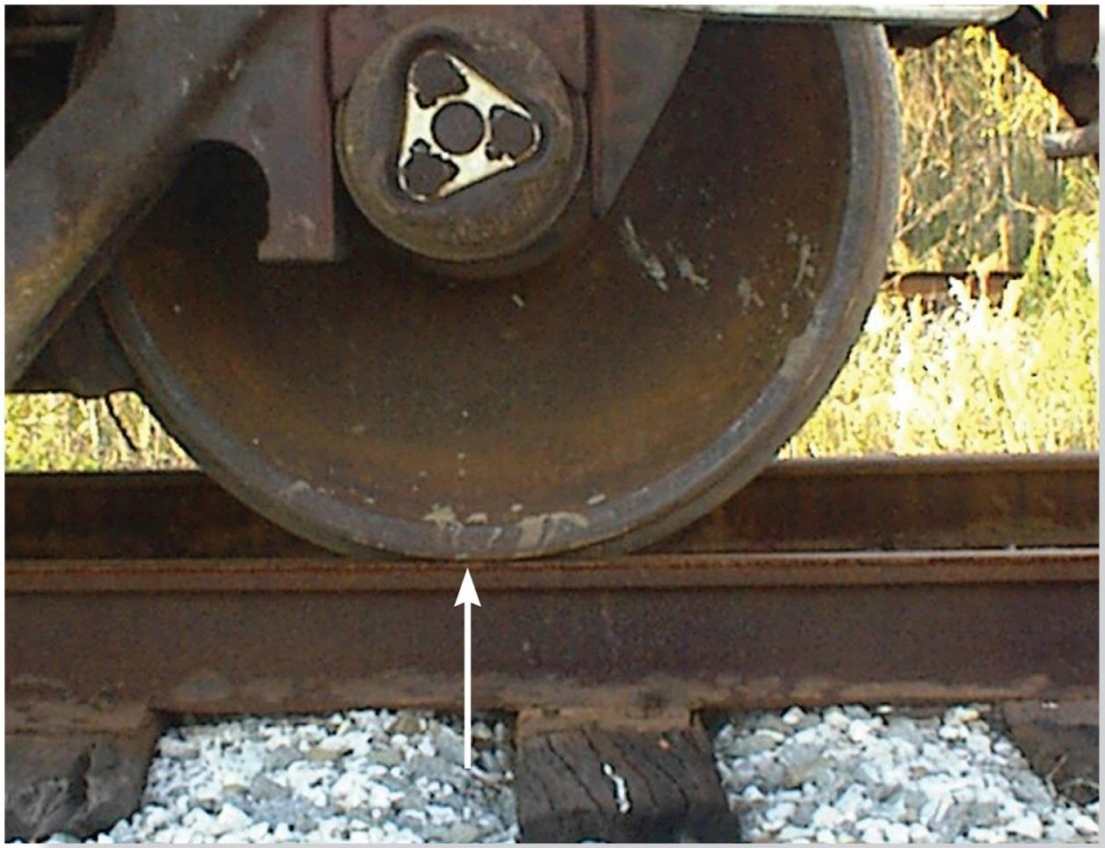

Figure: 01_PH002
Steel is a common engineering material that does not deform very much under load. Therefore, we can consider this railroad wheel to be a rigid body acted upon by the concentrated force of the rail.

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1.2 Fur

Idealizations

- **Concentrate**

  – Effect of loa
  on a body

  – Represente
  provided loadir
  overall size

  Eg: Contac
  ground

Figure: 01_PH001
Three forces act on the ring. Since these forces all meet at point, then for any force analysis, we can assume the ring to be represented as a particle.

1.2 Fundamentals Concepts

Newton's Three Laws of Motion

- **First Law**

  "A particle originally at rest, or moving in a straight line with constant velocity, will remain in this state provided that the particle is not subjected to an unbalanced force"

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1.2 Fundamentals Concepts

Newton's Three Laws of Motion

- **First Law**

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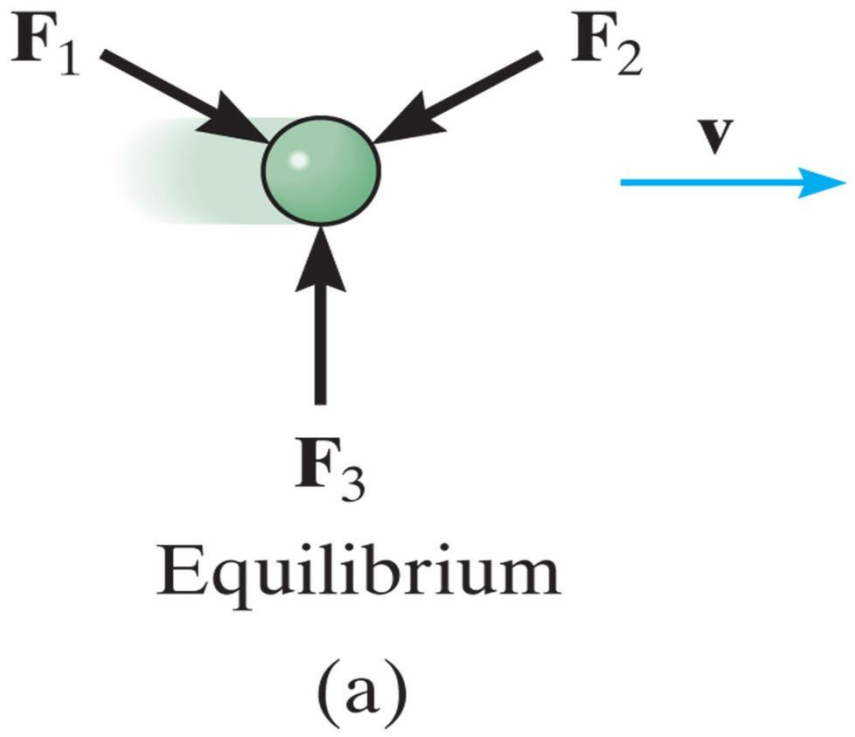

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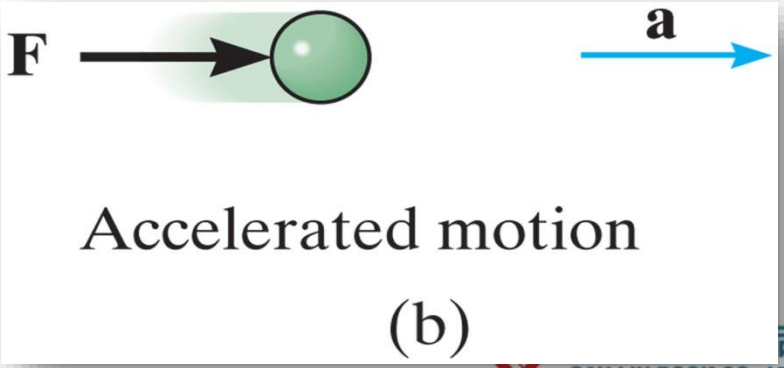
1.2 Fundamentals Concepts

Newton's Three Laws of Motion

- **Second Law**

  "A particle acted upon by an *unbalanced force* **F** experiences an acceleration **a** that has the same direction as the force and a magnitude that is directly proportional to the force"

$$F = ma$$



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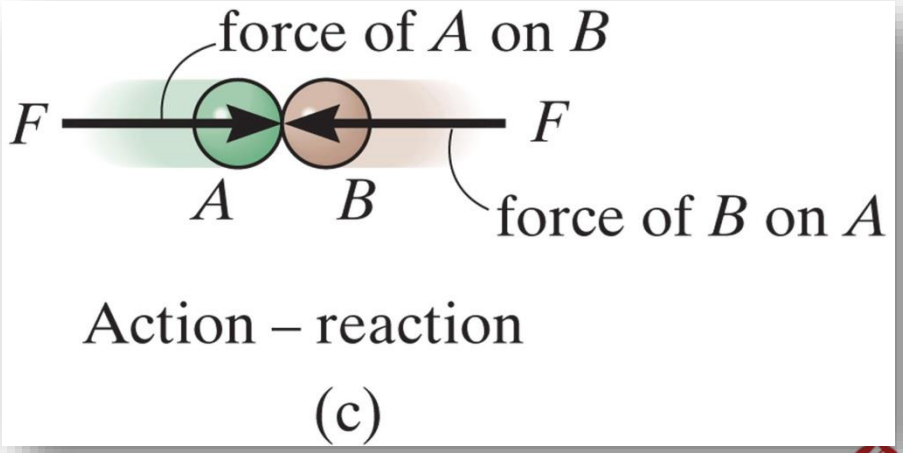
1.2 Fundamentals Concepts

Newton's Three Laws of Motion

- **Third Law**

  "The mutual forces of action and reaction between two particles are equal and, opposite and collinear"



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1.2 Fundamentals Concepts

Newton's Law of Gravitational Attraction

$$F = G\frac{m_1 m_2}{r^2}$$

F = force of gravitation between the two particles

G = universal constant of gravitation

m_1,m_2 = mass of each of the two particles

r = distance between the two particles

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1.2 Fundamentals Concepts

Weight $$W = G\frac{mM_e}{r^2}$$

Letting $g = GM_e / r^2$ yields

$$\boxed{W = mg}$$

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1.2 Fundamentals Concepts

Comparing F = mg with F = ma

- g is the acceleration due to gravity
- Since g is dependent on r, weight of a body is not an absolute quantity
- Magnitude is determined from where the measurement is taken
- For most engineering calculations, g is determined at sea level and at a latitude of 45°

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1.2 Funda

Comparing F =

- g is the acceleratio
- Since g is depend
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- Magnitude is dete
  measurement is ta
- For most enginee
  determined at sea

Figure: 01_PH003
The astronaut's weight is diminished, since she is far removed from the gravitational field of the earth.

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1.3 The International System of Units

- Length, time, mass and force
  - are not all independent from one another
  - related by Newton's second law of motion.

- $F = ma$ is maintained only if
  - Three of the units, called *base units*, are *defined*
  - Fourth unit is derived from the equation

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1.3 The International System of Units

SI Units

TABLE 1–1 International System of Units

Quantity	Length	Time	Mass	Force
SI Units	meter	second	kilogram	newton*
	m	s	kg	N $\left(\frac{kg \cdot m}{s^2}\right)$

*Derived unit.

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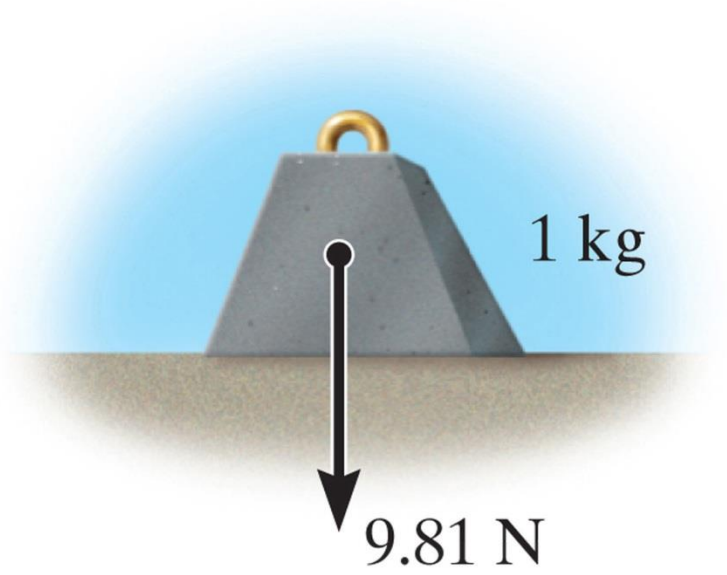
1.3 The International System of Units

- At the standard location,

  $g = 9.806\ 65\ m/s^2$

- For calculations, we use

  $g = 9.81\ m/s^2$



Thus,

$W = mg \qquad (g = 9.81 m/s^2)$

- Hence, a body of mass 1 kg has a weight of 9.81 N, a 2 kg body weighs 19.62 N

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1.3 The International System of Units

TABLE 1–2 Prefixes

	Exponential Form	Prefix	SI Symbol
Multiple			
1 000 000 000	10^9	giga	G
1 000 000	10^6	mega	M
1 000	10^3	kilo	k
Submultiple			
0.001	10^{-3}	milli	m
0.000 001	10^{-6}	micro	μ
0.000 000 001	10^{-9}	nano	n

*The kilogram is the only base unit that is defined with a prefix.

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1.3 The International System of Units

Rules for Use

- Quantities defined by several units which are multiples, are separated by a dot

  Eg: $N = kg.m/s^2 = kg.m.s^{-2}$

- The exponential power represented for a unit having a prefix refer to both the unit and its prefix

  Eg: $\mu N^2 = (\mu N)^2 = \mu N.\ \mu N$

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1.3 The International System of Units

Rules for Use

- With the exception of the base unit the kilogram, in general avoid the use of prefix in the denominator of composite units.

Eg: ~~N/mm~~ → kN/M ~~m/mg~~ → Mm/kg

- In calculations, represent numbers in terms of their base or derived units by converting all prefixes to powers of 10

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1.3 The International System of Units

Rules for Use

Eg: $(50\text{kN})(60\text{nm}) = [50(10^{3})\text{N}][60(10^{-9})\text{m}]$

$= 3000(10^{-6})\text{N.m}$

$= 3(10^{-3})\text{N.m}$

$= 3\ \text{mN.m}$

- The final result should be expressed using a <u>single prefix</u>

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1.4 Numerical Calculations

Dimensional Homogeneity

- Each term must be expressed in the same units

- All the terms of an equation can be combined if numerical values are substituted for the variables.

Keep in mind that problems in mechanics always involve the solution of dimensionally homogeneous equations, and so this fact can be used as a partial check for algebraic manipulations of an equation.

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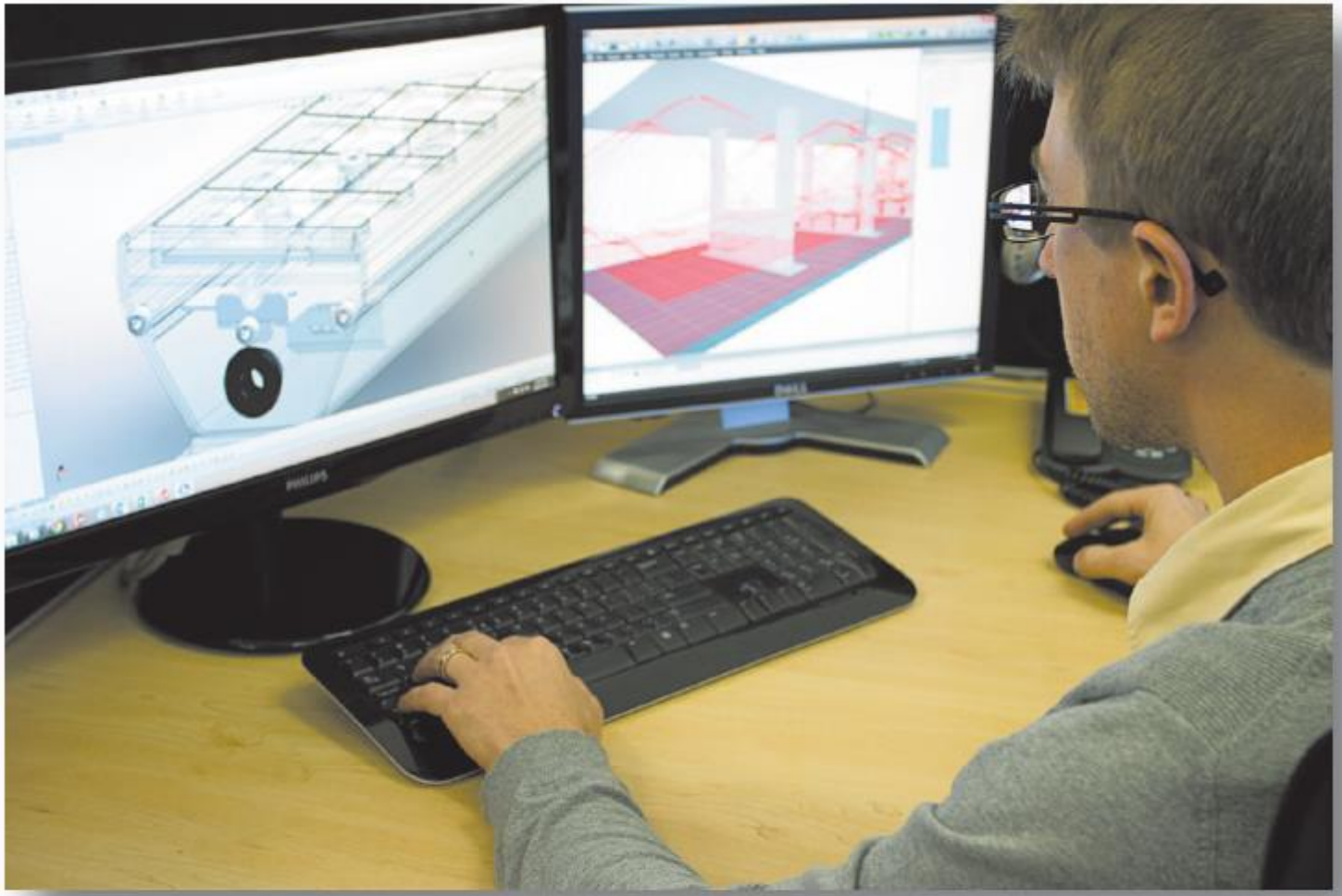

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Computers are often used in engineering for advanced design and analysis. (© Blaize Pascall/Alamy)

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1.4 Numerical Calculations

Engineering Notation

- If zero occur at the beginning of a number that is less than one, then the zeros are not significant.

Eg: 0.00821→ 3 significant figures : $8.21(10^{-3})$

Eg: 0,000582→ $0.582(10^{-3})$ or $582(10^{-6})$

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1.4 Numerical Calculations

Special case: end in “5”

Three significant figure

Even number 75.25 → 75.2

Odd number 0.1275 → 0.128

0.2555 → 0.256

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1.4 Numerical Calculations

Calculations

- Do not round off calculations until expressing the final result

- In this book, we will round off the answers to three significant figures

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1.5 General Procedure for Analysis

- Apply the relevant principles, generally in mathematical form. When writing any equations, be sure they are dimensionally homogeneous.
- Solve the necessary equations, and report the answer with no more than three significant figures.
- Study the answer with technical judgment and common sense to determine whether or not it seems reasonables.

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1.5 General Procedure for Analysis

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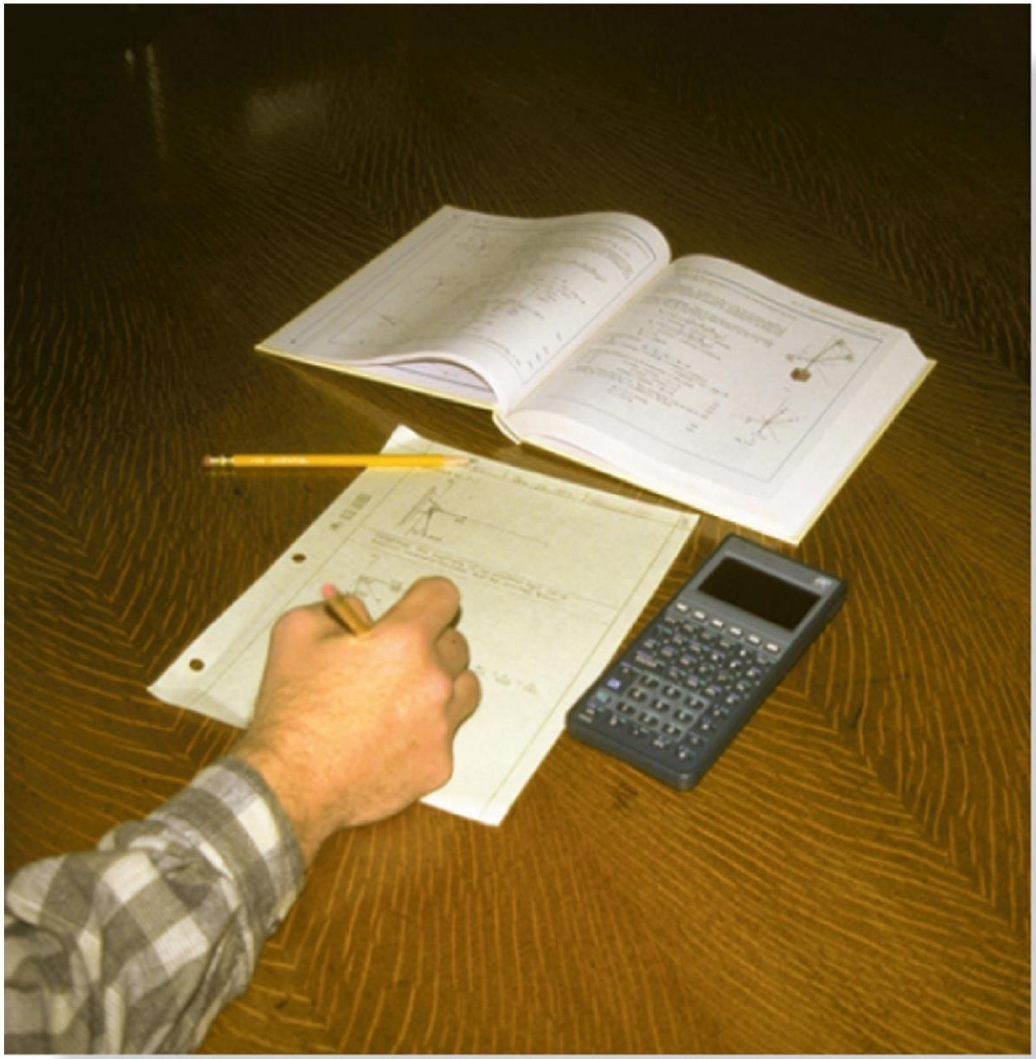

Figure: 01_PH005
When solving problems, do the work as neatly as possible. Being neat will stimulate clear and orderly thinking, and vice versa.

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EXAMPLE 1.1

Convert 100 km/h to m/s and 24 m/s to km/h.

SOLUTION

Since 1 km = 1000 m and 1 h = 3600 s, the factors of conversion are arranged in the following order, so that a cancellation of the units can be applied:

$$100\ \text{km/h} = \frac{100\ \cancel{\text{km}}}{\cancel{\text{h}}}\left(\frac{1000\ \text{m}}{\cancel{\text{km}}}\right)\left(\frac{1\ \cancel{\text{h}}}{3600\ \text{s}}\right)$$

$$= \frac{100(10^3)\ \text{m}}{3600\ \text{s}} = 27.8\ \text{m/s} \qquad \textit{Ans.}$$

$$24\ \text{m/s} = \left(\frac{24\ \cancel{\text{m}}}{\cancel{\text{s}}}\right)\left(\frac{1\ \text{km}}{1000\ \cancel{\text{m}}}\right)\left(\frac{3600\ \cancel{\text{s}}}{1\ \text{h}}\right)$$

$$= \frac{86.4\ (10^3)\ \text{km}}{1000\ \text{h}} = 86.4\ \text{km/h} \qquad \textit{Ans.}$$

NOTE: Remember to round off the final answer to three significant figures.

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EXAMPLE 1.2

Convert the density of steel 7.85 g/cm^3 to kg/m^3.

SOLUTION

Using 1 kg = 1000 g and 1 m = 100 cm, and arrange the conversion factor in such a way that g and cm^3 can be canceled out.

$$7.85\ \text{g/cm}^3 = \left(\frac{7.85\ \text{g}}{\text{cm}^3}\right)\left(\frac{1\ \text{kg}}{1000\ \text{g}}\right)\left(\frac{100\ \text{cm}}{1\ \text{m}}\right)^3$$

$$= \left(\frac{7.85\ \cancel{\text{g}}}{\cancel{\text{cm}^3}}\right)\left(\frac{1\ \text{kg}}{1000\ \cancel{\text{g}}}\right)\left(\frac{100^3\ \cancel{\text{cm}^3}}{1\ \text{m}^3}\right)$$

$$= 7.85(10^3)\ \text{kg/m}^3$$

Ans.

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EXAMPLE 1.3

Evaluate each of the following and express with SI units having an appropriate prefix: (a) (50 mN)(6 GN), (b) (400 mm)(0.6 MN)^2, (c) 45 MN^3/900 Gg.

SOLUTION

First convert each number to base units, perform the indicated operations, then choose an appropriate prefix.

Part (a)

$$
\begin{aligned}
(50\text{ mN})(6\text{ GN}) &= \left[50(10^{-3})\text{ N}\right]\left[6(10^{9})\text{ N}\right] \\
&= 300(10^{6})\text{ N}^2 \\
&= 300(10^{6})\,\cancel{\text{N}^2}\left(\frac{1\text{ kN}}{10^3\,\cancel{\text{N}}}\right)\left(\frac{1\text{ kN}}{10^3\,\cancel{\text{N}}}\right) \\
&= 300\text{ kN}^2 \qquad \textit{Ans.}
\end{aligned}
$$

NOTE: Keep in mind the convention $\text{kN}^2 = (\text{kN})^2 = 10^6\text{ N}^2$.

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EXAMPLE 1.3 CONTINUED

Part (b)

$$
\begin{aligned}
(400\text{ mm})(0.6\text{ MN})^2 &= \left[400(10^{-3})\text{ m}\right]\left[0.6(10^6)\text{ N}\right]^2 \\
&= \left[400(10^{-3})\text{ m}\right]\left[0.36(10^{12})\text{ N}^2\right] \\
&= 144(10^9)\text{ m}\cdot\text{N}^2 \\
&= 144\text{ Gm}\cdot\text{N}^2 \qquad \textit{Ans.}
\end{aligned}
$$

We can also write

$$
\begin{aligned}
144(10^9)\text{ m}\cdot\text{N}^2 &= 144(10^9)\text{ m}\cdot\cancel{\text{N}^2}\left(\frac{1\text{ MN}}{10^6\cancel{\text{N}}}\right)\left(\frac{1\text{ MN}}{10^6\cancel{\text{N}}}\right) \\
&= 0.144\text{ m}\cdot\text{MN}^2 \qquad \textit{Ans.}
\end{aligned}
$$

Part (c)

$$
\begin{aligned}
\frac{45\text{ MN}^3}{900\text{ Gg}} &= \frac{45(10^6\text{ N})^3}{900(10^6)\text{ kg}} \\
&= 50(10^9)\text{ N}^3/\text{kg} \\
&= 50(10^9)\cancel{\text{N}^3}\left(\frac{1\text{ kN}}{10^3\cancel{\text{N}}}\right)^3\frac{1}{\text{kg}} \\
&= 50\text{ kN}^3/\text{kg} \qquad \textit{Ans.}
\end{aligned}
$$

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